

CHARLES BENELLO

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EDUCATION

The University of Chicago

Master of Science in Financial Mathematics

- Coursework: ML in SWE, ML in Finance, Agentic AI, Distributed Systems, Reinforcement Learning

The University of Chicago | GPA: 3.75/4.0

Bachelor of Science in Mathematics and Computer Science, Minor in Korean

- Coursework: Database Systems, Operating Systems, Algorithms, Concurrency Control, Computer Architecture, Networks

EXPERIENCE

Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from **50% to 92–97%** by architecting a multi-stage pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial documents.
- Reduced hallucination rates across the platform's AI systems by building an **LLM-as-a-Judge evaluation framework** with automated benchmarks for faithfulness, numerical accuracy, and completeness.
- Cut **LLM inference costs** and improved response speed by optimizing context handling, adding personalization to client-facing AI, and removing context length barriers for large document ingestion.

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored **VLDB 2026** paper by designing a parallel external merge sort in **Rust** that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort.
- Sole-authored **HPTS 2026** paper by building a lightweight prediction model for sort configuration on cloud/disaggregated hardware, transferring calibration across machines within **3–5% of optimal**.
- Improved random access speed by **200%** and reduced memory footprint by **40%** by engineering Small String Optimization for storage structures.

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Designing **representation strategies** for transaction foundation models by comparing multi-token vs. single-token encoding and time-delta strategies (score-relative vs. inter-event) over **24M+ transactions** for fraud detection.
- Evaluating **BERT-style masked prediction vs. GPT-style next-token prediction** architectures with ablations across tokenization, time encoding, and context window size; built end-to-end data pipeline (CSV → Parquet → PyTorch).

Quantitative Researcher

Oct – Dec 2025

Bank of America (UChicago Project Lab)

Chicago, IL

- Built an AI-driven platform to **rebalance high-quality liquid asset portfolios**, generating trade recommendations that minimize risk exposure over 6-month projections under capital, liquidity, and net interest income constraints.
- Shipped **full-stack interfaces** (Next.js, Flask, React Native, FastAPI) and engineered a **multi-agent LLM debate system** where competing agents generate macro stress scenarios and produce probability-weighted reallocation recommendations.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab)

Lausanne, Switzerland

- Cut out-of-memory query times by **50–200%** by engineering a macroblock-based GPU decompression strategy in **CUDA/C++** for large analytical workloads.
- Drove a new **cost-based compression** research direction by benchmarking GPU compression techniques with resource contention models and building performance models enabling adaptive decompression on TPC-DS workloads.

PROJECTS

Transit Optimizer | Python | [cmbenello.github.io/covtsp](https://github.com/cmbenello/covtsp)

2026

- Beat the Guinness World Record by **60 min** (16h45m vs. 17h45m) by building a fully automated Covering TSP pipeline: given a city name, fetches GTFS feeds and **Google Maps** walking data, builds a time-expanded graph (160K nodes, 1.26M edges), and runs k-NN with urgency scoring, hard-station auto-detection, and epsilon-greedy search. Generalized to NYC (475 stations) and Berlin.

MiniMetro RL | Python, PyTorch | [cmbenello.github.io/MiniMetro](https://github.com/cmbenello/MiniMetro)

2025

- Built a custom **Gymnasium environment** simulating the transit-building game Mini Metro, with a full game engine (stations, lines, passengers, routing) and trained a **PPO agent** to learn network construction strategies.

Relational Database Engine | Rust | github.com/cmbenello/CrustyDB

2023

- Implemented a **multi-threaded relational database from scratch**: B+ tree indexing, concurrent buffer pool management with page-level locking, heap storage, and a query execution engine supporting SQL operations.

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Go, TypeScript, JavaScript, SQL, CUDA

Frameworks & Tools: PyTorch, React, Next.js, Flask, FastAPI, React Native, Git, Linux, DuckDB, XGBoost

Concepts: Distributed Systems, Database Internals, Algorithms, Data Structures, Concurrency, GPU Programming, ML, System Design

PUBLICATIONS

[1] R. Otaki*, **C. Benello***, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (*Equal contribution)

[2] **C. Benello**. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026*.

[3] R. Otaki, J.H. Chang, **C. Benello**, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

ADDITIONAL

Languages: Korean (Intermediate), Italian (Intermediate), French (Basic)

Interests: Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting